

DATASHEET

Prithvi R

12/05/2026

1 Motivation

The aim of this datasheet is to clearly provide descriptions of the black-box functions in the BBO capstone challenge and associated `.csv` files which contain their evaluations throughout the challenge.

2 Collection & Composition

There are eight black box functions in total. Each function has a set of known evaluations $(\mathbf{x}, y)$ associated with it. The dimensionality of each black-box function $f(\mathbf{x}) = y$ is determined from the length of the input vector $\mathbf{x}$. At the start of the challenge, a separate optimiser was instantiated for each black-box function $f_i(\mathbf{x})$. Each optimiser was warm-started by supplying all known evaluations to its surrogate model. Following this, the first query $n = 1$ was generated to initiate the optimisation loop. To obtain an evaluation $(\mathbf{x}, y)$, the query $\mathbf{x}$ was submitted to an online portal managed by Imperial. During the challenge, the optimisers for each function generated sample points for new evaluations. We refer to these evaluations as generated evaluations. At each round n , the evaluation of the previous round $(n - 1)$ was fed into the optimiser for the generation of a new query (sample point in the search space). This process continued until the evaluation budget was spent. See the table below for a description of each function involved in the challenge:

Func.	Dim.	Known Eval.	Gen. Eval.	Description
1	2	10	14	$f(\mathbf{x})$ describes radiation field intensity with spatial coordinates (x_1, x_2) .
2	2	10	13	Log-likelihood emulation $l(\theta)$ with parameters (x_1, x_2) .
3	3	15	14	Drug discovery model where x_1 , x_2 and x_3 define compound ratios and $y < 0$ encodes side-effects.
4	4	30	14	ML warehouse allocation model returning deviation from baseline cost.
5	4	20	13	Industrial yield function with unimodal response surface.
6	5	20	13	Cake ingredient optimisation with negative scoring objective.
7	6	30	13	Deep learning hyperparameter performance metric.
8	8	40	13	Deep learning validation accuracy over 8 hyperparameters.

Table 1: The dimensionality of the input $\mathbf{x}$ (Dim.), the number of known evaluations, generated evaluations and description of each black-box function in the challenge.

Both the known and generated evaluations for each black-box function are saved in a human-readable CSV format with the name scheme `func_i_evals.csv` where i denotes the function number. There is only a single table in each respective `.csv` file. The fields of this table are `Query Number`, `x1`, `x2`, etc. and Each row in the table an represents an evaluation. Known evaluations have negative integer query numbers (-1, -2, -3, etc.) while generated evaluations have positive integer ones (1, 2, 3, etc.). The data files have been created with the intention of being loaded via the `pandas` nodule. An example of the boilerplate code:

```
# Importing
import pandas as pd

# Reading evaluations
df = pd.read_csv("dataset/func_1_evals.csv")
df
```

Out[1]:

	Query Number	x1	x2	f(X)
0	-1	0.319404	0.762959	1.322677e-79
1	-2	0.574329	0.879898	1.033078e-46
2	-3	0.731024	0.733000	7.710875e-16
3	-4	0.840353	0.264732	3.341771e-124
4	-5	0.650114	0.681526	-3.606063e-03
5	-6	0.410437	0.147554	-2.159249e-54
6	-7	0.312691	0.078723	-2.089093e-91
7	-8	0.683418	0.861057	2.535001e-40
8	-9	0.082507	0.403488	3.606771e-81
9	-10	0.883890	0.582254	6.229856e-48
10	1	0.732787	0.733181	3.589724e-16
11	2	0.500000	0.450000	-5.847155e-06
12	3	0.716104	0.731080	2.491766e-13
13	4	0.701749	0.728584	2.669673e-11
14	5	0.697745	0.727874	8.202194e-11

Figure 1: Reading the .csv containing the evaluations for the first $i = 1$ black-box function using **pandas** on a Jupyter notebook.

3 Preprocessing

No preprocessing steps have been performed on the output of each query y presented in this dataset. However, some preprocessing steps were taken when supplying each optimiser with evaluations during the optimisation loop. Mostly importantly, the optimisers were supplied $(\mathbf{x}, -y)$ because **skopt**, the Python package used, assumes the minimisation of an objective function $f(\mathbf{x})$. Therefore by minimising $-f(\mathbf{x})$, $f(\mathbf{x}) = y$ was maximised. For more details of preprocessing steps specific to each black-box function, see their corresponding .ipynb notebook.

4 Maintenance

The .csv files storing the evaluations of each black-box function can be found in the **dataset** folder of this repository. Note that this dataset (and datasheet) will be maintained until the termination of the BBO challenge.